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MILITARY SPECIFICATION

SIGNALS, ILLUMINATION, GROUND, STAR, CLUSTER (WHITE, GREEN, AMBER, AND RED) LOADING, ASSEMBLING, AND PACKING

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for one type of cluster star ground signals, designated as Signals, Illumination, Ground, Star, Cluster M18A2, M20A2, M22A2, M52A2, M18A2B2, M20A2B2, M22A2B2, and M52A2B2.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-223 — Powder, Black.

JAN-P-663 — Powder, Fuze.

MIL-P-2550 — General Specification for Ammunition except Small Arms Ammunition.

MIL-P-3207 — Propellant, Powders, for Small Arms Ammunition, Caliber .45.

MIL-P-20334 — Pyrotechnics, Specification for White and Colored Composition.

MIL-P-20464 — Pyrotechnics: Methods of Determination of Candlepower and Color Value.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection of Attributes.

DRAWINGS

ORDNANCE CORPS

78-0-38 — Signal, Ground, Star, Cluster (for Use with Grenade Launcher, M7 and M8) Assembly.

78-0-55 — Signal, Ground, Star, Cluster (for Use with Grenade Launcher, M7 and M8) Assembly.

FSC 1370

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76–1–1143 — Ammunition Containers, Fiber M184A1 for Ground Signal. Assembly, Details, Packing and Marking.

76–1–1144 — Ammunition Container, Fiber, M113A1 for Ground Signal. Assembly, Details, Packing and Marking.

20–4–613 — Marking Diagram and Sealing for Wooden Packing Boxes.

75–14–629 — Marking and Sealing for Ammunition Fiber Containers.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 Signals. The signals shall comply with all requirements specified on Drawings (dwg.) 78–0–38 or 78–0–55, as applicable, and with all requirements specified in applicable specifications.

3.3 Moisture content (see 6.1).

3.3.1 *Star composition.* The moisture content of the star composition, at the site of loading at the time of loading the signals, shall not exceed 0.1 percent when determined as specified in 4.4.1.

3.3.2 *Fuze composition.* The mosture content of the fuze composition, at the site of loading at the time of loading the signals, shall not exceed 0.3 percent when determined as specified in 4.5.1.

3.3.3 *Black powder.* The moisture content of the balck powder, at the site of loading at the time of loading the signals, shall not exceed 0.3 percent when determined as specified in 4.6.

3.3.4 *Smokeless powder.* The moisture content of the smokeless powder, at the site of loading at the time of loading the signals, shall not exceed 0.3 percent when determined as specified in 4.7.1.

3.3.5 *Felt.* The moisture content of the felt at the site of loading at the time of loading the signals, shall not exceed 6 percent when determined as specified in 4.8.

3.4 Functioning. A score of not less than 2750 shall be obtained when tested as specified in 4.10.

3.5 Visual examination. Visual examination at the time of loading shall show no moisture on the surface of the metal parts.

3.6 Workmanship. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used. The moisture content of the loaded item shall be kept to a minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot. A lot shall consist of all signals loaded and assembled by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Each lot shall contain:

(a) black cartridges from not more than one lot.

(b) signals of not more than one designation.

(c) black powder from not more than one lot.

(d) smokeless powder from not more than one lot.

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The inspection lot for non-destructive inspection within the above limitations, shall be designated by the inspector.

4.1.1 When specified in the contract or order (see 6.1) groups for shipment shall be formed from each lot. The number of units in a shipment group and the marking shall be as specified.

4.2 Sampling.

4.2.1 *For non-destructive inspection.* Sampling for non-destructive inspection shall be conducted in accordance with Standard MIL–STD–105.

4.2.2 *For static test of illuminant assembly.* Samples shall be selected for test in accordance with table 1.

TABLE I

Lot size	Sample size	k
1 to 22,000	20	2.25
22,001 to 110,000	35	2.39
110,001 and over	60	2.50

4.2.3 *For functioning test.* Thirty signals shall be selected from each lot for test.

4.3 Inspection.

4.3.1 *General.* Inspection shall be as specified in Specification MIL–G–2550 and as specified herein.

4.3.2 *Components.* It shall be ascertained that prior to assembly all components of subassemblies of the signals procured under separate specifications or drawings have been inspected, tested, and accepted in accordance with their respective specifications or drawings.

4.3.3 *Loading and assembling.* The preparation and loading of the fuze composition, the loading of the star composition, the black powder, the smokeless powder, and the assembling of the felt shall be observed and inspected for conformance to this specification by a Government inspector at the place of manufacture.

4.3.4 *Packing and marking.* It shall be ascertained that the packing and marking of the signals conform to this specification.

4.3.5 *Classification of defects.* Unless otherwise specified in the contract or order, inspection of the signals shall be as follows:

4.3.5.1 *Housing, fuze (see dwg. 78–2–308 covering a detail of 78–0–38 and dwg. 78–2–331 covering a detail of 78–0–55).*

Categories and defects	*Method of inspection*
Major *AQL 0.25 percent*	
1. Intermediate charge hole obstructed	Visual
Minor	
None	

4.3.5.2 *Stabilizer assembly (see dwg. 78–2–310 covering a detail of 78–0–38 and 78–0–55).*

Categories and defects	*Method of inspection*
Major *AQL 0.25 percent*	
1. Inside diameter of stabilizer tube minimum (min.)	Gage
Minor *AQL 4.00 percent*	
41. Pitch diameter of thread, maximum (max.)	Gage
42. Minor diameter of thread max.	Gage
43. Length of perfect thread, min.	Gage
44. Length to thread, max.	Gage
45. Depth of threaded cavity	Gage
46. Wall thickness of tube at shoulder min.	Gage
47. Outside diameter of shoulder min.	Gage
48. Outside diameter of fin	Gage
49. Length of taper	Gage

4.3.5.3 Assembly, prior to assembling *identification top and loading propelling charge (see dwg. 78–0–38, and 78–0–55).*

Categories and defects	*Method of inspection*
Critical *AQL 0.015 percent*	
A. Color of star assembly incorrect	Visual

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Categories and defects	*Method of inspection*
Major — *AQL 1.00 percent*	
1. Cushioning wad missing	Visual
2. Star assembly missing	Visual
3. Quickmatch missing	Visual
4. Priming charge missing from two or more star assemblies	Visual
5. Three or more holes missing from cup	Visual
6. Three or more holes missing from disc	Visual
7. Wad missing	Visual
8. Powder pellet missing	Visual
9. Inner charge missing	Visual
10. Igniting charge missing	Visual
11. Ring seal missing or loose	Visual
Minor — *AQL 4.00 percent*	
41. Pitch diameter of thread, min.	Gage
42. Major diameter of thread, min.	Gage
43. Length of perfect thread, min.	Gage
44. Length of flange	Gage
45. Disc inadequately crimped in cup	Visual
46. Quickmatch and star quickmatch ends not tied	Visual
47. Retaining disc missing	Visual
48. Priming charge missing from one star assembly	Visual
49. Less than three holes missing from cup	Visual
50. Less than three holes missing from disc	Visual

4.3.5.4 *Assembly, after loading propelling charge (see dwg. 78–0–38 and 78–0–55).*

Categories and defects	*Method of inspection*
Critical — *AQL 0.015 percent*	
A. Propelling charge missing	Visual
Major	
None	
Minor	
None	

4.3.5.5 *Assembly, as taken from unsealed fiber container (see dwg. 78–0–38 and 78–0–55).*

Categories and defects	*Method of inspection*
Critical — *AQL 0.015 percent*	
A. Color marking misleading	Visual
B. Propelling charge escaping	Visual
Major — *AQL 1.00 percent*	
1. Misalignment between body and stabilizer tube, max.	Gage
2. Body loose	Manual
3. Metal defective	Visual
4. Tape twisted between plug and stabilizer tube	Visual
5. Fin damaged	Visual
6. Identification top damaged	Visual
7. Rust on interior of stabilizer tube	Visual
8. Color marking unidentifiable	Visual
Minor — *AQL 2.50 percent*	
41. Length to fin from end of stabilizer tube, max.	Gage
42. Identification top not fully seated	Visual
43. Identification top inadequately sealed	Visual
44. Weld missing	Visual
45. Protective coating inadequate	Visual
46. Nomenclature marking misleading or unidentifiable	Visual

4.3.5.6 *Sealed fiber container, as taken from unsealed wood packing box (see dwg. 76–1–1143, 76–1–1144, and 75–14–629).*

Categories and defects	*Method of inspection*
Major — *AQL 0.40 percent*	
1. Container cut or damaged through all impregnated layers	Visual
2. Tape incomplete or badly wrinkled	Visual
3. Metal end loose or distorted	Visual
Minor — *AQL 1.00 percent*	
41. Gap between cover and body of container excessive	Visual-tactile
42. Contents loose	Manual
43. Tear tab length incorrect	Visual
44. Color of tape incorrect	Visual
45. Marking misleading or unidentifiable	Visual

4.3.5.7 *Sealed wood packing box (see dwg. 76–16–232, covering a detail of 78–0–38, dwg. 76–16–354 covering a detail of 78–0–55, and 20–4–613.)*

Categories and defects	*Method of inspection*
Major — *AQL 0.65 percent*	
1. Box damaged	Visual
2. Hardware broken or loose	Visual-Manual
3. Strapping broken or loose	Visual-Manual
4. FSC symbol misleading or unidentifiable	Visual

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Minor	*AQL 1.00 percent*
41. Hardware improperly engaged	Visual-Manual
42. Strapping improperly assembled	Visual-Manual
43. Car seal unsealed or improperly positioned	Visual-Manual
44. Contents loose	Manual
45. Handle insecure	Manual
46. Marking misleading or unidentifiable	Visual

4.4 Test of star composition.

4.4.1 *Moisture content.* Five samples of star composition from each lot shall be selected for determination of the moisture conetnt. An accurately weighed portion of approximately 10 grams (gms.) of the sample star composition shall be placed in a tared weighing dish approximately 2 inches in diameter. The dish shall be covered and the dish and contents weighed. The cover shall be removed and the dish and contents heated at 70 degrees plus or minus 2 degrees centigrade (70±2°C.) for 2 hours. The cover shall be replaced, the dish and contents cooled in a desiccator, and weighed. The loss in weight shall be calculated to percentage of moisture in the sample. The lot shall be rejected if any of the samples exceed the requirement of 3.3.1.

4.4.2 *Ingredients (see* 6.1*).* Five samples of star composition from each lot shall be selected for analysis of the ingredients. The ingredients of the star composition shall be determined in accordance with Specification MIL–P–20334, to determine compliance with the requirements specified on the drawing. The lot shall be rejected if any of the samples exceed these requirements.

4.5 Test of fuze composition.

4.5.1 *Moisture content.* Five samples of fuze composition from each lot shall be selected for determination of the moisture content. The method of determining the percentage thereof shall be in accordance with Specification JAN–P–663. The lot shall be rejected if any of the samples exceed the requirements of 3.3.2.

4.5.2 *Ingredients (see 6.1).* Five samples of fuze composition from each lot shall be selected for analysis of the ingredients. The ingredients of the fuze composition shall be determined in accordance with Specification JAN–P–663 to determine compliance with the requirements specified on the drawing. The lot shall be rejected if any of the samples, exceed these requirements.

4.6 Determination of moisture content of black powder. Five samples of black powder from each lot shall be selected for determination of moisture content. The method of determining the percenage thereof shall be in accordance with Specification JAN–P–223. The lot shall be rejected if any of the samples exceed the requirements of 3.3.3.

4.7 Test of smokeless powder.

4.7.1 *Moisture content.* Five samples of smokeless powder from each lot shall be selected for determination of the moisture content. A known weight of approximately three gms. of the sample smokeless powder shall be transferred to a tared weighing dish or covered watch glass. The sample shall be dried for 2 hours at 70°C. The dish shall be covered, cooled in a desiccator and weighed. The loss in weight shall be calculated as percent of moisture in the sample. The lot shall be rejected if any of the samples exceed the requirements of 3.3.4.

4.7.2 *Ingredients (see 6.1).* Five samples of smokeless powder from each lot shall be selected for analysis of the ingredients. The ingredients of the smokeless powder shall be determined in accordance with Specification MIL–P–3207 to determine compliance with the requirements on the drawing. The lot shall be rejected if any of the samples exceed these requierments.

4.8 Determination of moisture content of felt. Five samples of felt from each lot shall be selected for determination of the moisture content. The sample shall be cut into small

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squares of approximately ¼ inch. The accurately weighed portion of approximately 50 gms. of the sample shall be transferred to a balloon flask of suitable fuze, and a sufficient volume of trichloroethylene shall be added so that the flask shall be about half full. The graduated portion of the moisture tube (see figure 1) shall be filled with trichloroethylene and attached to the flask and condenser of the Allehn type. A drying tube containing anhydrous calcium chloride, or other suitable desiccant, shall be connected to the top of the condenser to keep out atmospheric moisture. The flask shall be heated by any suitable means so that the distillate falls from the end of the condenser at the rate of two or three drops per second. The distillation shall be continued until the water layer in the measuring tube remains constant for three successive readings taken at 30 minute intervals. The assembled flask, moisture tube and condenser shall be removed from the source of heat and the condenser shall be washed down with small portions 5 milliliters (ml.) of fresh trichloroethylene until the volume of water layer becomes constant and no droplets of water are observed in the condenser. This usually requires 25 to 30 ml. trichloroethylene. The point of content of the top meniscus with the wall of the tube and the highest point of the lower meniscus shall be read. The difference in these readings shall be determined and recorded as the volume of water found.

The percentage of moisture in the sample shall be calculated as follows:

$$\text{Percentage of total moisture} = \frac{100\ V}{W}$$

where:

V = volume of water, ml.
W = weight of sample, gms.

The lot shall be rejected if any of the samples exceed the requirements of 3.3.5.

4.9 Static test of illuminant assembly.

4.9.1 *Procedure.* One star shall be removed from the representative signals and tested by supporting the star in a horizontal position with the axis of the star perpendicular to the photometric axis, and ignited by means of quickmatch so that star burns flame up. The total burning time, candlepower, and color value shall be measured and recorded to determine compliance with the burning time, candlepower, and color value requirement specified on the drawing. Determination of burning time, candlepower, and color value shall be made in accordance with Specification MIL–P–20464.

4.9.2 *Rejection.* If the average candlepower minus "K" times the standard deviation is less than the lower limit of the applicable drawing requirement or if the burning time of any illuminant assembly is less than the applicable drawing requirements, or if the color value is less than the applicable drawing requirement, the inspection lot shall be rejected without further test.

4.10 Functioning test. The following test shall be performed at a Government proving ground.

4.10.1 *Procedure.* Immerse the representative signals in water at 70 degrees plus or minus 10 degrees Fahrenheit (70±10°F.) for 2 hours to a depth of not less than 6 nor more than 9 inches. Remove the signals from the water and wipe dry the exterior surfaces. Fire the signals from a grenade launcher, M7 without retaining, spring, mounted on a United States Rifle M1, with the rifle in vertical position and butt of rifle resting on normal solid ground, or in a suitable mounting which will give approximately the same altitude of functioning; within 1 hour after removal from water. Record the time from firing to functioning of the signal. Do not fire the signals when wind velocity exceeds 15 miles per hour. Observe functioning and score in accordance with table 11. Give each signal a score of 100 minus such deductions as are listed. Whenever two types of failures occur in the same signal, deduct only the major penalty.

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TABLE II. *Method of scoring*

	Deduct
1. Signal functions satisfactorily	0
2. One star fails to function	25
3. Two stars fail to function	50
4. Three or more stars fail to function	100
5. Burning time of cluster is less than 4 seconds, is greater than 10 seconds, or extinction intervals exceed 3 seconds	50
6. Signal bursts on a launcher or burst at an altitude of less than 150 feet[1]	300
7. Signal functions at an altitude of 150 to 500 feet (Provided the total number of failures of the type does not exceed 4 for each 30 signals tested)	0
8. For each failure of type (7) in excess of 4 for each 30 signals tested	50
9. Stabilized assembly bursts or ruptured	150
10. Signal fails to function	100
11. Color of any star other than required	475
12. Average height of functioning of all signals functioning at an altitude of over 150 feet is less than 550 feet but not less than 500 feet	275
13. Average height of functioning of all signals functioning at an altitude of over 150 feet and less than 500 feet	475

4.10.2 *Retest and rejection.* If a score of less than 2750 but not less than 2550 is obtained, the lot shall be considered eligible for retest, using 60 signals. If, at any time, a score of less than 2550 is obtained, or if, at any time, a combined total of the scores of the original test plus the retest is less than 8250, the lot shall be rejected without further test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, labeling, and marking. Packaging, packing, labeling, and marking shall be specified on Drawing 76–1–1143 or 76–1–1144, Drawing 76–16–232 covering a detail of Drawing 78–0–38 or Drawing 76–16–354 covering a detail of 78–0–55, and Drawing 20–4–613.

5.2 Data cards. Data card information shall be as specified in Specification MIL–G–2550.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Signal required.

(c) Certification required to guarantee that the moisture content requirements specified in 3.3 and that the analysis of ingredients specified in 4.4.2, 4.5.2, and 4.7.2 have been complied with.

(d) When required, size of group for shipment and marking (see 4.1.1).

6.2 This specification is intended to supersede Picatinny Arsenal Purchase Description PA–PD–585, 30 July 1954.

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[1] In case of premature functioning if observation of the altitude of functioning is questionable, but initial flight is normal, a signal will be considered as functioning above 150 feet, if the time for firing to functioning is .75 second or more, and penalty under 4.10.1–6, will not be applied. However the signals may be penalized, if applicable, under other parts of the scoring table.

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